

Evidence-First VLM Cascades for Embedded Work-Zone State Estimation: The Role of Model-Specific Temporal Calibration

Anonymous Authors
Paper under double-blind review

Abstract—Work-zone traversal state (OUTSIDE, APPROACHING, INSIDE, EXITING) is usually estimated with an object detector, semantic verification and temporal smoothing. We ask how a single INT4 vision-language model (VLM) on an embedded Jetson AGX Orin compares with such a pipeline when every system is measured on equal terms. Our *evidence-first cascade* operates one 2B VLM through asymmetric-cost prompt channels and hysteresis, under a *latency-honest* protocol where each system’s measured latency determines frame dropping. On 208 held-out ROADWork videos it ties a ported detector pipeline on macro-F1 and event recall but trails on precision, nuisance rate and compute. The temporal constants matter more than the model. Inherited from the 2B, they leave the 4B reasoner of a world model at macro-F1 0.290; recalibrated, it reaches 0.515. Recalibrating the detector pipeline’s own state machine raises it from 0.470 to 0.551, the best result we measure. After calibration no VLM-based system beats it, and a counted hidden-Markov filter on the detector score alone matches it with counted tables instead of a threshold search. These conclusions hold on drive-disjoint folds, and they argue for reporting each system’s calibration and comparing at matched nuisance.

I. INTRODUCTION

Work zones concentrate difficult driving-perception conditions: their appearance is regional and unstandardized, geometry changes daily, evidence such as cones, barrels, signs, and workers is small or occluded, and both misses and nuisance alerts are costly. They accounted for roughly 101,000 U.S. crashes and 899 fatalities in 2023 [1], yet production ADAS rarely model work zones explicitly and temporary speed limits are often absent from digital maps.

The strongest onboard approach combines a YOLO-style work-zone detector, CLIP verification [2], exponential smoothing, and hysteresis to estimate four traversal states [3], [4]. It is fast and effective after a zone is established, but reported IoU falls from 0.56/0.47 on OUTSIDE/INSIDE to 0.11/0.09 on APPROACHING/EXITING [3]. These boundaries are precisely where an advisory is most useful: an alert after the vehicle reaches the barrels has little anticipatory value.

Meanwhile, fine-tuned VLMs describe work-zone scenes well offline [5], and weight-only quantization plus embedded runtimes bring small VLMs to automotive hardware [6], [7]. We ask whether a single fine-tuned VLM, operated through cooperating prompt channels, can estimate traversal state in real time and compete with the detector pipeline under identical conditions.

Under an identical embedded protocol, the detector-free VLM is competitive on state coverage but trails on precision, nuisance rate, and compute. The comparison itself, however, rests on temporal constants. Inherited from one checkpoint, they leave a 4B world model at INSIDE IoU 0.007; recalibrated, it reaches 0.574. The same calibration applied to the detector pipeline’s hand-tuned state machine raises its macro-F1 from 0.470 to 0.551. Once every system is calibrated on the same data, no VLM-based system beats the recalibrated detector, a counted filter on the detector score alone matches it without a search, and these conclusions hold on drive-disjoint folds.

At ~ 10 ms per output token, a description costs ~ 500 ms and a two-token answer 75 ms, so our cascade lets evidence quality gate smoothing (Section III-B): a transcribed sign can bypass a vote while a binary answer cannot.

Our contributions are three. **(1) Evaluation:** a latency-honest protocol in which measured cycle time drives frame dropping, checked against live concurrent execution, with per-model recalibration by evidence replay. **(2) Findings:** temporal constants do not transfer between the systems we test, and recalibrating the detector baseline changes the ranking: the joint estimator beats the hand-tuned pipeline (+0.076) but not the calibrated one, and no VLM-based system does; this holds on drive-disjoint folds, at matched false alarms and with matched search budgets. **(3) Systems:** an evidence-first cascade that runs a quantized 2B VLM in real time on the Orin, an INT4 deployment of the Cosmos3-Edge reasoner on the same device, and a counted hidden-Markov filter that matches the recalibrated baseline from the detector score alone, with an analysis that locates the residual VLM gap in ego-relevance.

II. RELATED WORK

a) Work-zone perception.: Early systems used structured environmental models with sensor fusion [8] or frame-level classification [9]. ROADWork [5] contains 4,375 videos and 9.6K annotated images across 18 U.S. cities, and shows that open-vocabulary models improve sharply after in-domain fine-tuning (+32.2 AP for detection and +36.7 SPICE for description). The closest system [3] deploys a ROADWork-trained YOLO detector with CLIP verification, EMA smoothing, and hysteresis, and defines the traversal-state evaluation we adopt.

b) VLMs and world models for driving.: Large VLMs support end-to-end planning [10], [11], open-loop descrip-

tion [12], and reasoning [13], and Cosmos world models plan at datacenter or automotive-SoC scale [14], [15]. We instead estimate one safety-relevant scene state under an embedded latency budget.

c) *Efficient inference and cascades.*: AWQ [6] and embedded runtimes [7] enable multi-Hz VLM inference on embedded GPUs. Detection cascades [16] and LLM cascades [17] chain models of increasing cost and escalate on scalar confidence. Our stages share one model: prompts and token budgets create the cost asymmetry, and evidence semantics (transcribed text versus a binary answer) drive escalation.

d) *Temporal filtering and fair comparison.*: Our joint estimator is a discrete hidden Markov model [18] run as a Bayes forward filter [19], with tables estimated by counting; we claim no novelty for the filter. That baselines need the same tuning budget as the proposed method is known [20]; we measure how much it changes conclusions for embedded temporal estimators.

III. SYSTEM

A. Two embedded deployments

We deploy *two* models on the same Jetson AGX Orin 64 GB, which shares its SoC with the automotive DRIVE AGX Orin, adapted in the two ways such a model admits: one by task fine-tuning, one by quantization and per-model calibration alone. Both run the identical prompt channels, cascade, protocol and splits, so every comparison in this paper varies the model and its adaptation, never the pipeline around it.

Fine-tuned 2B. A 2B-parameter VLM (Qwen3-VL architecture [21], initialized from Cosmos-Reason2-2B [22]), used for perception only. We fine-tune it through a ROADWork-derived curriculum [5]: work-zone VQA ($\sim 46K$ pairs), ego-state supervision, sign reading, a general-driving retention mix, and $\sim 10K$ work-zone-free BDD100K hard negatives. Before the last stage, the existence gate answered “yes” on 95% of clean scenes; after it, 0% on external negative controls (clean BDD100K scenes; on ROADWork OUTSIDE seconds, which include visible but irrelevant work, it is far less selective, Table IV).

The language backbone is quantized to INT4 (W4A16 AWQ, group size 128), the vision tower stays FP16, and both are compiled to TensorRT engines on-device [7] (1.38 GB language + 0.83 GB vision). Decoding costs 10.4 ms per output token and dominates, followed by vision encoding (44.6 ms at 720p, ~ 20 ms at the 480px we use).

Quantized 4B world model. Cosmos3-Edge (C3E) [14] is a 4B physical-AI world model pairing an autoregressive reasoner with a trajectory-generating diffusion transformer. We use only the reasoner; the diffusion transformer is never run, and “world model” below refers to this reasoner. We quantize it to W4A16 AWQ INT4 (1.27 GB) and keep its visual tower in FP16 (983 MB), both as TensorRT engines on the same Orin; its frozen 26×46 patch grid fixes inputs at 736×416 , and channel latencies are 162/182/488/556 ms (GATE/EGO/DESC/SIGN). It receives no work-zone training data and has never seen this benchmark. Building its engines

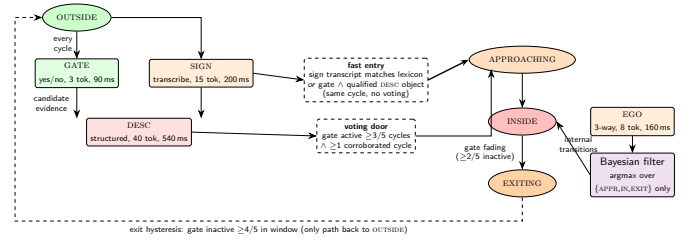


Fig. 1. Evidence-first cascade. In OUTSIDE, GATE and SIGN run every cycle and candidate evidence invokes DESC. Entry requires a sustained, corroborated vote or high-confidence same-cycle evidence. In active states, EGO drives a Bayesian filter; sustained gate clearance returns to OUTSIDE, while partial fading triggers EXITING. Constants are calibrated per model: 5/3/4/2 for the 2B and 3/3/2/1 for C3E, so the structure transfers but the numbers do not. Diagram values are the 2B’s, with rounded latencies (measured values in Section V-D).

required six patches to the embedded runtime; the patch set and deployment recipe are released with the code.

B. Prompt channels and their failure modes

We operate four fixed-prompt channels with fixed token budgets. Because the first generated token is an answer-start token, the minimum useful budget is two tokens.

a) *Affirmation bias.*: The GATE asks a neutral existence question and is calibrated on clean scenes, but binary prompts cannot be specialized by rephrasing. Variants asking whether work affects the vehicle’s path all answered “yes” on OUTSIDE probes, while distance prompts answered “close” regardless of scene. The learned discrimination resides in the trained question; prompt nuance does not transfer. Binary outputs are therefore weak evidence requiring a temporal vote, never sole triggers.

b) *Faithful transcription.*: The SIGN channel returns stable, verbatim text such as “SHOULDER WORK” and correctly reports absence. A binary rephrasing misses signs that transcription catches on the same frames. We therefore transcribe and derive a binary value by matching a work-zone keyword lexicon.

c) *Structured corroboration.*: The DESC channel’s object–location prefix often verbalizes disqualifying context—e.g., “barriers on right sidewalk” or “work vehicle parked.” Corroboration therefore requires a work-zone object without such a qualifier. Prompts, budgets and lexicons are released with the code.

C. Evidence-first cascade

The cascade (Fig. 1) allocates compute by state.

Outside (patrol). GATE and SIGN run every cycle (2B at 480px: p95 259 ms, 3.9 Hz, Section V-D); either can invoke DESC. Entry into APPROACHING has two paths: a voting path, with the gate active in at least K_{enter} of the last N cycles and one corroboration, or fast entry on a lexicon-matching sign or gate+DESC agreement in the same cycle. Fast entry avoids a voting delay of up to N cycles (~ 1.3 s for the 2B) for evidence such as a transcribed “ROAD WORK AHEAD.”

Active states. EGO updates a Bayesian filter whose argmax is restricted to the three active states. Otherwise, a single “outside” reply immediately after sign-triggered entry would erase the sign’s anticipatory value. Only sustained gate clearance returns the system to OUTSIDE; partial fading triggers EXITING, which EGO never produced even when offered explicitly. DESC refreshes every 1.5 s. For the 2B, the thresholds ($N=5$, $K_{\text{enter}}=3$, $K_{\text{exit}}=4$, $K_{\text{fade}}=2$) were fixed on calibration data; C3E gets its own (Section VI). With EGO disabled, as selected for C3E, the filter’s emission is uniform: APPROACHING becomes INSIDE on the transition prior alone, 13 cycles after entry, and EXITING and OUTSIDE still come from the fade and clearance votes.

D. Joint estimator over both evidence streams

Combining the detector and the world model by grafting one onto the other’s temporal machinery does not work. The ported detector counts cycles (a 150-cycle maximum approach duration, 20 minimum frames outside) tuned for its native loop; the cascade votes over a window tuned for the 2B’s ~ 260 ms cycle, while C3E answers only every ~ 1.5 s alongside the detector. Reusing either as the spine imports constants that are wrong for the combination, the same problem as inheriting thresholds across models (Section VI), one level up.

The joint estimator therefore inherits no constant from either pipeline: it has no EMA, no hysteresis threshold and no cycle counter, and its tables are counted; its remaining design choices (bucket edges and the evidence rate τ below) were set on calibration data (Section VII). It is a discrete hidden Markov model over the four states, run as a forward filter [18], with maximum-likelihood tables. Its observation combines the detector’s calibrated frame score (six buckets), the world model’s gate, sign-keyword and corroboration answers ordered by evidence strength (six levels), and the age of that answer (three buckets), for 108 discrete cells. The emission $P(\text{observation} \mid \text{state})$ is counted once per detector cycle on the calibration split with Laplace smoothing, and it is joint rather than factorized because the interaction is the signal: the world model’s APPROACHING is reliable when the detector corroborates it and mostly a false alarm when it does not (73% of those frames are OUTSIDE). The transition is a per-second rate matrix, counted as ground-truth transitions over dwell time and scaled by the elapsed dt of each step, so it does not depend on the loop rate. Finally, the log-likelihood is scaled by dt/τ with $\tau = 0.5$ s: a world-model answer persists across roughly 30 detector frames, and applying its likelihood once per frame treats 30 correlated looks as independent, which drove an early version to 135 false alarms per hour.

At run time the detector runs at its native rate and is never blocked. The world model is polled asynchronously and re-polled as soon as it answers; its last answer is held, with its age as part of the observation. In our runs both models execute in one process, each charged its own measured latency on the simulated clock; we record the raw evidence with these clocks once per video and replay the filter on it, so ages and

frame drops in every reported result are the measured ones. Section VII checks this replay against a live concurrent run.

IV. EVALUATION PROTOCOL

a) *Benchmark.*: We use the detector baseline’s 520-video ROADWork-derived traversal benchmark [3]: Boston and Seattle footage at 1920×1080 , 30 fps, with 484K interval-labeled frames. A state-stratified 60/40 video split assigns all design choices and thresholds to calibration (312 videos, 2.59 h) and final results to validation (208 videos; 192,135 frames, 1.78 h, 13 pure-negative videos); the benchmark totals 4.37 h of annotated driving. All systems see the same split. All 25 drives occur in both partitions, so they are not route-independent; Section VII-A therefore repeats every replay-fitted comparison with drive-disjoint folds. Fine-tuning images and benchmark videos both derive from ROADWork, a condition the ROADWork-trained baseline shares.

b) *Latency-honest camera simulation.*: Embedded perception cannot pause the world. After each measured inference cycle of duration Δ on the target hardware, the replay advances by Δ and drops intervening frames exactly as a live camera would. Slower systems therefore observe fewer, sparser frames and pay latency inside the accuracy result rather than beside it. Predictions are held between cycles and scored per frame.

c) *Metrics.*: We adopt the baseline’s suite [3]: frame accuracy and macro-F1; per-state IoU; event precision/recall, where an interval is correct if it overlaps one of the same state; matched-event entry offset (Entry MAE: mean absolute offset over all states; IN entry med.: median absolute offset of INSIDE entries); and false alarms per hour, the rate of predicted active intervals overlapping no annotated active interval. Videos beginning in a state contribute no entry event. We keep the baseline’s overlap rule for comparability and add a lead metric for timing (Section V-A). Event metrics capture whether a warning occurs, IoU and timing how well it is placed. Confidence intervals are paired 95% cluster-bootstrap intervals (10,000 replications) that resample drives, so that correlated videos of one drive are drawn together; video-level intervals are reported where they differ in sign.

V. RESULTS: ONE FINE-TUNED MODEL VS. THE DETECTOR

A. Comparison with the deployed detector system

Table I places every system on the same videos, device and protocol. This section compares the fine-tuned 2B with the ported detector pipeline as published; the recalibrated detector and C3E are the subject of Section VI, the counted filter and joint estimator of Section VII. With greedy decoding, our default (Section V-C), the fine-tuned 2B is statistically tied with the detector pipeline on frame accuracy and macro-F1 (-0.018 , $[-0.041, +0.003]$) and matches its INSIDE event recall, but trails on event precision (56.8% vs. 79.9%) and false alarms (56.2 vs. 30.9 per hour). This is a competitive detector-free result rather than a replacement result. A causal 1 s INSIDE debounce raises precision to 76.0% (macro-F1 0.478) without retraining; Section VII-A compares all systems at matched

TABLE I

VALIDATION, 208 VIDEOS, ONE PROTOCOL AND DEVICE (%; TIMING IN S). DET.+CLIP: HAND-TUNED [3] AND RECALIBRATED (SECTION VI); DET. *cnt.*: DETECTOR SCORE IN THE COUNTED FILTER; 2B: GREEDY, NO POST-FILTER (RECALIBRATED: 0.459); JOINT: DETECTOR+C3E IN THE COUNTED FILTER. LEAD: INSIDE ONSETS PRECEDED BY ≥ 2 S OF ACTIVE STATE. F1@20: AT ≤ 20 FALSE ALARMS/H (SECTION VII-A).

Metric	Det.+CLIP		Det. <i>cnt.</i>	2B	C3E	Joint
	hand	cal.				
Frame acc.	62.9%	64.8%	68.2%	62.0%	65.5%	66.1%
Macro F1	0.470	0.551	0.524	0.453	0.515	0.546
IoU OUT	0.617	0.627	0.652	0.581	0.615	0.641
IoU APPR	0.124	0.296	0.219	0.152	0.278	0.246
IoU IN	0.544	0.552	0.573	0.532	0.574	0.578
IoU EXIT	0.106	0.153	0.124	0.062	0.071	0.160
IN ev. recall	95.7	77.8	88.6	97.3	87.0	85.9
IN ev. prec.	79.9	92.5	75.9	56.8	71.5	66.8
False alarms/h	30.9	20.2	22.5	56.2	64.6	25.9
Entry MAE	2.52	2.31	2.44	2.52	2.15	2.22
IN entry med.	3.47	2.10	2.73	2.67	2.50	2.27
Lead ≥ 2 s	79.4	78.8	73.1	76.2	86.2	66.9
F1@20	0.410	0.529	0.520	0.409	0.487	0.537

false alarms. Entry MAE is equal (2.52 s), and the 2B’s INSIDE entry median is better by 0.80 s.

The four-state pattern matters more than aggregate accuracy. Both pipelines perform well on established INSIDE intervals but poorly on EXITING. The 2B’s APPROACHING IoU is higher (0.152 vs. 0.124), as text and distant semantic evidence appear before detector counts stabilize; Section VI shows that the detector closes this gap once its own thresholds are recalibrated.

B. Operational false-alarm characterization

Over 1.78 h, false active-state events occur at 56.2/h with greedy decoding, versus 30.9/h for the detector pipeline. Specificity is measured on 75,918 OUTSIDE frames (70.3% correctly held; detector 75.6%) and on the 13 pure-negative validation videos, where the 2B remains OUTSIDE for 63.8% of their frames (detector 51.2%). Residual errors are predominantly the ego-relevance family of Section VIII, not clean-scene noise. Of 156 matched INSIDE entries, 80.8% fire early (median 2.9 s; detector 92.8%, 3.6 s), which an advisory prefers but which stretches intervals. Reducing nuisance further is a calibration question, addressed at matched false alarms in Section VII-A.

C. Ablations and design validation

Table II toggles one component at a time from the sampled-decoding cascade. Five of the seven change macro-F1 by at most 0.02; the exceptions are the sign channel and the checkpoint. The sign channel must transcribe rather than classify: an equal-cost binary question reduces accuracy from 62.5% to 52.5% and raises false alarms fivefold. Greedy decoding costs 0.5 accuracy points but improves precision and alarms, so we use it as the reproducible default. The qualifier filter adds 2.2 accuracy points and removes 9.6 alarms per hour. Fast entry buys anticipation: 10 more points of onsets are warned ≥ 2 s

TABLE II

SINGLE-TOGGLE ABLATIONS OF THE 2B CASCADE (VALIDATION, 208 VIDEOS). LEAD: INSIDE ONSETS PRECEDED BY ≥ 2 S OF ACTIVE STATE.

Variant	Acc.	F1	IN prec.	FA/h	Lead
Cascade, sampled	62.5	0.464	54.1	60.1	79.4
greedy decoding	62.0	0.453	56.8	56.2	76.2
binary SIGN	52.5	0.407	46.1	305.8	86.2
no fast entry	63.1	0.462	54.3	34.3	69.4
no qualifier filter	60.3	0.447	53.9	69.7	81.2
no SIGN channel	63.2	0.456	53.4	47.8	82.5
unrestricted filter	61.4	0.445	53.1	69.1	73.8
base checkpoint	48.8	0.289	–	29.2	–

ahead, at 26 more alarms per hour. The base checkpoint under the 2B’s thresholds collapses (Section VI).

D. Operational cost

At 480px, p50/p95 latency is 83/90 ms (GATE), 136/149 ms (EGO), 155/169 ms (SIGN), and 468/508 ms (DESC). Patrol and active cycles run at p95 259 ms (3.9 Hz) and 239 ms (4.2 Hz); the heaviest path is 768 ms (1.3 Hz). Here “real time” means keeping pace with a live camera at these rates (at 30 m/s, a 259 ms cycle is 7.8 m of travel), not meeting a specific alert deadline. Power comes from tegrastats rail readings in two sessions with different idle draw, so we compare increments over idle. The VLM draws 36.5 W median (42.5 W peak) over a 7.9 W idle, i.e. 28.6 W above idle, and 3–26 J per cycle. In a later session (11.1 W idle) the detector alone drew 33.5 W, 22.4 W above idle, at 24 ms and 0.8 J per cycle. The VLM thus costs about $1.3\times$ the power above idle and 4–33 $\times$ the energy per decision.

VI. VARYING THE MODEL: CALIBRATION AND GENERALIZATION

Section V confounded each checkpoint with thresholds fitted to the 2B. We separate the two by recalibrating every system on its own evidence, which changes two conclusions above.

a) *Temporal hyper-parameters do not transfer.*: With the 2B thresholds, C3E scores macro-F1 0.290 and INSIDE IoU 0.007 at 3.8% recall. This is an artifact, not an inability to recognize being inside a zone. Recalibrating the same four integers for C3E moves these to 0.515, 0.574, and 87.0% on validation: INSIDE IoU rises 0.567 absolute and recall 83.2 points, with model, prompts and videos fixed. The search covers $N \in \{3, 5, 7, 9\}$, $K_* \in 1..N$ and the EGO/SIGN switches (4,896 configurations), scored by macro-F1 on the 100 calibration videos with recorded VLM evidence. The selected configuration ($N=3$, $K_{\text{enter}}=3$, $K_{\text{exit}}=2$, $K_{\text{fade}}=1$; calibration macro-F1 0.484) is shorter and less hysteretic, consistent with less noisy per-frame evidence, and disables EGO (Section VIII). Because it sat on the smallest N of the grid, we later extended the grid to $N \in \{1, 2, 4\}$; the optimum did not change. Recording each channel’s output once per sampled frame and replaying the state machine turns a threshold sweep from GPU-hours per configuration into CPU-seconds.

The identical search on the identical videos moves the fine-tuned 2B by only +0.006 on validation (0.453→0.459), against +0.225 for C3E: a search that merely rewarded flexibility would have helped both models equally.

b) The baseline too.: The argument applies to the detector pipeline, whose ported state machine keeps constants hand-tuned for its native loop on a different split. We recorded every component on every cycle of all 520 videos (detector counts before and after CLIP verification, the global CLIP score and the colour-context score) and replay the complete pipeline, keeping the CLIP and colour weights at the app’s values; replaying the hand-tuned configuration reproduces the live run (0.473 vs. 0.470). We searched its thresholds, EMA rate, minimum-outside count and maximum approach duration on the 312 calibration videos (three grids of 4,800 configurations), selecting on calibration macro-F1 alone. Validation macro-F1 rises to 0.551 (+0.077, [+0.039, +0.108]; 0.486→0.538 on calibration) with 92.5% INSIDE precision and 20.2 false alarms per hour (Table I), the strongest system we measure. The same search on the detector score alone, without CLIP, gives 0.435→0.530.

The same procedure also measures fine-tuning under matched conditions. Recalibrating the *base* checkpoint—identical family, scale, architecture and visual tower, differing only in the work-zone curriculum—lifts it from macro-F1 0.281 to 0.383 on the calibration videos used for the search, against 0.451 to 0.496 for the fine-tuned version. Given their own thresholds, fine-tuning is worth +0.113 macro-F1 and +8.8 accuracy points, not the near-total collapse (INSIDE IoU 0.535→0.012) that the base checkpoint shows under the fine-tuned model’s thresholds. It buys a real and specific capability in domain.

Every decision in this paper is made on the 312-video calibration partition and none on validation: prompts, budgets and the qualifier lexicon on the full partition; VLM thresholds by replaying evidence recorded on a 100-video subset of it; the detector’s thresholds and the counted filter’s tables on all 312 videos. The 208 validation videos enter only as the reported result. Search budgets are therefore unequal, and they favour the detector: three grids over 312 videos against one over 100. Given the VLM’s budget, one grid on the same 100 videos, the recalibrated detector+CLIP pipeline reaches 0.522 on validation, still above its hand-tuned version (+0.052, [+0.038, +0.070]) and level with C3E (−0.008 for C3E, [−0.050, +0.029]).

c) No configuration dominates.: Calibrated, the world model and the detector pipeline are statistically tied on macro-F1 (−0.036, [−0.080, +0.005]) and fail complementarily: C3E leads INSIDE IoU (0.574 vs. 0.552) and entry MAE, the recalibrated detector INSIDE precision and nuisance (20.2 vs. 64.6 false alarms/h). Section VII combines their evidence.

VII. COMBINING THE DETECTOR AND THE WORLD MODEL

The joint estimator of Section III-D reaches macro-F1 0.546 at 25.9 false alarms per hour (Table I). Against the

hand-tuned detector+CLIP pipeline it gains 0.076 macro-F1 ([+0.038, +0.104]); against the recalibrated pipeline the two are tied (−0.005, [−0.046, +0.033]), as are their INSIDE IoU and false alarms, and the joint estimator trades INSIDE event precision (66.8% vs. 92.5%) for recall (85.9% vs. 77.8%). Against the calibrated world model it gains 0.031 macro-F1 ([+0.006, +0.058]) and 0.089 EXITING IoU and removes 38.8 alarms per hour. We treat the comparisons against the recalibrated detector as confirmatory and the per-metric differences as exploratory.

Every table is counted on all 312 calibration videos, with no selection among fitting protocols; 5-fold cross-validation within calibration gives macro-F1 0.535 (s.d. 0.019) at 30.3 false alarms/h. The constants that are not counted were set on calibration data: τ trades accuracy for nuisance (cross-validated: 0.25, 0.538 at 44.7/h; 0.5, 0.535 at 30.3/h; 1.0, 0.504 at 22.0/h; 2.0, 0.444 at 16.8/h), and we kept the knee at 0.5. The tables are sparse but concentrated: 68 of the 108 cells occur in calibration and 20 carry 97% of the observations; with 3 or 10 detector buckets instead of 6, validation macro-F1 is 0.538 and 0.545.

a) Concurrent execution.: In the recorded runs behind every reported result, the detector cycle took p50 44 ms (p95 59 ms) and the world model answered every 1.54 s. To check the replay, we ran both models truly concurrently on the Orin, in wall-clock time from a decoded camera stream, on 20 validation videos, twice. Sharing the GPU raises the detector cycle from 24 ms alone to 54 ms (p95 59 ms, 19 Hz), while the world model answers every 1.44 s. Run live, the detector-only counted filter reproduces its replay on these videos (macro-F1 0.425 vs. 0.423); the joint estimator scores 0.44 in both live runs against 0.50 in replay, a paired difference (−0.07, [−0.16, +0.01]) that 20 videos cannot resolve. The Orin draws 39.0 W median (46.9 W peak) with both models, against 33.5 W for the detector alone and 11.1 W idle.

b) Where the gain comes from.: A channel ablation refits the filter without each stream. Removing the world model entirely, including the age of its answer, costs 0.022 macro-F1 and lowers the alarm rate (25.9 to 22.5 per hour). Removing the detector costs 0.086 macro-F1 and collapses EXITING IoU from 0.160 to 0.017. Of the world-model channels only DESC pays for itself in domain (−0.016 without it); SIGN changes nothing, and removing GATE raises macro-F1 to 0.558 but adds 6 alarms per hour. The detector score alone inside the counted filter scores 0.524 at 22.5 false alarms per hour, tied with the recalibrated detector+CLIP pipeline on both (−0.026, [−0.062, +0.009]; +2.2/h, [−8.1, +11.1]) without CLIP or a threshold search, and with 11.2 fewer alarms per hour than the state machine recalibrated on the same score ([−24.6, −2.1]). On this benchmark the most efficient option is therefore the detector inside the counted filter. The world model’s evidence is complementary where the detector’s is weak: on the 10 validation onsets preceded by a transcribed work-zone sign, the joint estimator and C3E warn ≥ 2 s ahead on all ten, against eight for the recalibrated detector+CLIP pipeline.

TABLE III

DRIVE-DISJOINT 5-FOLD REPLAY, ALL 520 VIDEOS: EVERY SYSTEM FITTED WITHOUT THE TEST DRIVES. F1@20: MACRO-F1 AT ≤ 20 FALSE ALARMS PER HOUR UNDER A SHARED PERSISTENCE FILTER.

System	F1	IoU IN	IN prec.	FA/h	F1@20
Det.+CLIP, hand-tuned	0.481	0.570	79.8	26.1	0.417
Det.+CLIP, recalibrated	0.535	0.560	90.7	18.7	0.535
Det. score, hand-tuned	0.434	0.546	73.3	47.0	0.410
Det. score, recalibrated	0.523	0.543	86.7	43.9	0.420
Det., counted filter	0.524	0.590	73.5	22.5	0.520
Det.+C3E, counted filter	0.537	0.591	70.8	28.7	0.523

c) *A learned alternative.*: A strictly causal three-stage TCN [23] trained on the same evidence ties the best macro-F1 (0.550) but at 71.9 false alarms per hour.

A. Drive-disjoint folds and matched false alarms

To rule out route-specific calibration, we pool all 520 videos and split the 25 drives into five folds of about 104 videos. Each replay-fitted system is refitted on the other four folds’ drives only, each detector state machine with the full three-grid search, and replayed on the held-out fold, so no test video shares a drive with any video used to fit it (Table III). The conclusions of Sections VI and VII hold: recalibration raises the detector+CLIP pipeline by 0.054 macro-F1 ($[+0.026, +0.078]$, drive-level) and the state machine on the detector score alone by 0.089 ($[+0.058, +0.114]$); the counted filters tie the recalibrated pipeline ($-0.010, [-0.036, +0.017]$, on the detector score alone; $+0.002$ with the world model), the detector-only filter at a similar alarm rate (22.5 vs. 18.7/h, $[-1.7, +8.8]$). Restricted to the 208 validation videos, drive-disjoint fitting changes each system’s macro-F1 by at most 0.010 from Table I.

Macro-F1 compares systems at different operating points, from 18 to 65 false alarms per hour. We therefore pass every system through the same causal persistence filter, which shows an active interval only after it has lasted d seconds, and increase d until false alarms fall to 20/h (last column of Tables I and III). At matched nuisance, the recalibrated and counted systems lead: on drive-disjoint folds the recalibrated detector+CLIP pipeline keeps 0.535 at $d=0$, the joint estimator 0.523 and the counted detector filter 0.520 at $d \leq 0.75$ s, against 0.42 for the hand-tuned pipeline and for the state machine recalibrated on the detector score alone. Matching nuisance also changes the anticipation ranking on the validation split: at ≤ 30 false alarms per hour, C3E warns ≥ 2 s before 58% of INSIDE onsets and the 2B before 54%, against 73% for the counted detector filter and 79% for the recalibrated detector+CLIP pipeline.

VIII. FAILURE ANALYSIS: THE EGO-RELEVANCE GAP

Nearly all residual false-positive mass has one signature: descriptions are factually correct about objects irrelevant to the vehicle’s path. Examples include a parked work vehicle more than 60 m away, sidewalk reconstruction behind barriers, and residual detour signs for another street. The qualifier filter

TABLE IV

GATE “YES” RATE BY TRUE STATE, INSIDE-VS-OUTSIDE BALANCED ACCURACY OF THE BINARY ANSWER (BA), AND DESC CORROBORATION ON OUTSIDE; SAME 100 CALIBRATION VIDEOS, 1 HZ.

Checkpoint	GATE “yes” when truth is					
	OUT	APPR	IN	EXIT	BA	DESC
2B base	3.8%	53.1%	71.8%	9.8%	0.84	31.3%
2B fine-tuned	42.4%	85.3%	92.2%	51.2%	0.75	10.7%
C3E zero-shot	29.4%	93.3%	94.0%	34.4%	0.82	35.7%

catches only cases where the model verbalizes this context; it does not emit qualifiers reliably, and binary or distance prompts do not recover the judgment.

a) *Response bias.*: The three checkpoints were recorded on the same labeled videos, which separates how often a model says “yes” from how well its answer tracks the state (Table IV). The base 2B rarely answers yes outside a zone and its gate separates INSIDE from OUTSIDE best (balanced accuracy 0.84); fine-tuning moves the same gate to 42.4% and 0.75. ROADWork’s OUTSIDE includes visible but irrelevant work, so a “yes” there is often correct perception and missing ego-relevance. The fine-tuned model’s advantage under matched calibration therefore does not come from its gate: its description corroborates on 10.7% of OUTSIDE seconds against 31.3% for the base model.

Incomplete fine-tuning coverage is not the whole explanation. C3E transcribes signs correctly and emits plausible 2D boxes, yet labels most verified INSIDE frames “off-road/sidewalk” in ego-lane probes, nearly independently of ground truth.

Perceiving objects and judging whether they claim the ego path are distinct capabilities, and none of the prompts we tried bridged them in either model, across two scales and two adaptation strategies; the gap is not specific to one fine-tuning run. It also explains weak EXITING IoU for every language-only system: receding objects remain visible. The joint estimator reaches 0.160 on EXITING only through the detector: without it, EXITING IoU falls to 0.017. The detector’s object score decays as objects recede, and the counted filter reads that decay directly; no language channel supplies it.

Trajectory-grounded supervision is the natural remedy. ROADWork’s 129K path annotations [5], unused in our fine-tuning, can derive ego-relevance labels from recorded trajectories rather than a lexicon.

IX. LIMITATIONS

Our claims are bounded in four ways. (i) *Data*: fine-tuning and evaluation both use ROADWork, which covers two daytime U.S. cities; the drive-disjoint folds of Section VII-A address the replay-fitted temporal models, and the VLMs were not re-fine-tuned per fold. (ii) *Baseline*: every comparison uses our port of the detector pipeline, measured on our split and protocol (0.470; 0.600 published on the original split). (iii) *Search*: reported configurations are the best found on finite grids; the conclusions hold with matched budgets (Section VI).

(iv) *Scope*: replay is open-loop, and we claim computational feasibility, not safety readiness.

X. CONCLUSION

An INT4 2B VLM can estimate work-zone state at live-camera rates, matching detector event recall under a latency-honest paired comparison but not its precision or nuisance rate. Testing the assumptions behind that comparison changes its conclusion. Inherited temporal thresholds move INSIDE IoU by 0.567 absolute, so cross-model comparisons must recalibrate. The same calibration raises the detector pipeline from 0.470 to 0.551 macro-F1, after which no VLM-based system beats it, and a counted hidden-Markov filter on the detector score alone matches it without a threshold search; this holds on drive-disjoint folds. Comparisons of embedded temporal estimators should therefore report how each system, the baseline included, was calibrated, and compare at matched nuisance. The residual ego-relevance gap resisted every prompt we tried in both models and calls for trajectory-grounded supervision. Upon acceptance we release the replay harness, prompts, metrics and baseline port so that both latency-honest comparison and per-model calibration are reproducible.

REFERENCES

- [1] National Highway Traffic Safety Administration, “2023 fatality analysis reporting system (FARS) annual report file,” U.S. Department of Transportation, accessed via workzonesafety.org, 2023.
- [2] A. Radford *et al.*, “Learning transferable visual models from natural language supervision,” in *Proceedings of the International Conference on Machine Learning (ICML)*, 2021, pp. 8748–8763.
- [3] A. Martinez-Sanchez *et al.*, “ZONEWISE: Zone worksite inference with speed estimation,” in *Proceedings of the 28th International Technical Conference on the Enhanced Safety of Vehicles (ESV)*, Toronto, Canada, 2026, first place, ESV 2026 student competition.
- [4] —, “Vision-language work zone intelligence for safety-critical speed regulation of mixed-autonomy vehicles in dynamic environments,” *arXiv preprint arXiv:2606.08860*, 2026.
- [5] A. Ghosh *et al.*, “ROADWork: A dataset and benchmark for learning to recognize, observe, analyze and drive through work zones,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, pp. 6132–6142.
- [6] J. Lin *et al.*, “AWQ: Activation-aware weight quantization for on-device LLM compression and acceleration,” in *Proceedings of Machine Learning and Systems (MLSys)*, 2024.
- [7] NVIDIA, “TensorRT Edge-LLM: High-performance LLM and VLM inference runtime for embedded platforms,” <https://github.com/NVIDIA/TensorRT-Edge-LLM>, 2026.
- [8] W. Shi and R. R. Rajkumar, “Work zone detection for autonomous vehicles,” in *Proceedings of the IEEE International Intelligent Transportation Systems Conference (ITSC)*, 2021, pp. 1585–1591.
- [9] F. Abodo *et al.*, “Detecting work zones in SHRP 2 NDS videos using deep learning-based computer vision,” in *Proceedings of the IEEE International Conference on Machine Learning and Applications (ICMLA)*, 2018, pp. 679–686.
- [10] J.-J. Hwang *et al.*, “EMMA: End-to-end multimodal model for autonomous driving,” *arXiv preprint arXiv:2410.23262*, 2024.
- [11] X. Tian *et al.*, “DriveVLM: The convergence of autonomous driving and large vision-language models,” in *Conference on Robot Learning (CoRL)*, 2024.
- [12] A.-M. Marcu *et al.*, “LingoQA: Visual question answering for autonomous driving,” in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024.
- [13] NVIDIA, “Cosmos-reason1: From physical common sense to embodied reasoning,” *arXiv preprint arXiv:2503.15558*, 2025.
- [14] —, “Cosmos 3: Omnimodal world models for physical ai,” *arXiv preprint arXiv:2606.02800*, 2026.
- [15] —, “Alpamayo-r1: Bridging reasoning and action prediction for generalizable autonomous driving in the long tail,” *arXiv preprint arXiv:2511.00088*, 2025.
- [16] P. Viola and M. Jones, “Rapid object detection using a boosted cascade of simple features,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2001.
- [17] L. Chen, M. Zaharia, and J. Zou, “FrugalGPT: How to use large language models while reducing cost and improving performance,” *arXiv preprint arXiv:2305.05176*, 2023.
- [18] L. R. Rabiner, “A tutorial on hidden Markov models and selected applications in speech recognition,” *Proceedings of the IEEE*, vol. 77, no. 2, pp. 257–286, 1989.
- [19] S. Thrun, W. Burgard, and D. Fox, *Probabilistic Robotics*. MIT Press, 2005.
- [20] G. Melis, C. Dyer, and P. Blunsom, “On the state of the art of evaluation in neural language models,” in *International Conference on Learning Representations (ICLR)*, 2018.
- [21] Qwen Team, “Qwen3-VL: A versatile vision-language model series,” <https://github.com/QwenLM/Qwen3-VL>, 2025.
- [22] NVIDIA, “Cosmos-Reason2-2B model card,” <https://huggingface.co/nvidia/Cosmos-Reason2-2B>, 2025.
- [23] Y. A. Farha and J. Gall, “MS-TCN: Multi-stage temporal convolutional network for action segmentation,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 3575–3584.